MM7025 - COMPUTER ALGEBRA, COMPUTER EXERCISES 3

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(1) Let $f_i = x^i + y^i + z^i + u^i + 1$ for $i = 1, \ldots, 5$ be elements in $\mathbb{C}[x, y, z, u]$.
- Determine whether the system of equations $f_1 = f_2 = f_3 = f_4 = 0$ has a solution or not, that is, check if $\mathrm{V}(\langle f_1, f_2, f_3, f_4\rangle)$ is non-empty or not.
- Determine whether the system of equations $f_1 = f_2 = f_3 = f_4 = f_5 = 0$ has a solution or not, that is, check if $\mathrm{V}(\langle f_1, f_2, f_3, f_4, f_5\rangle)$ is non-empty or not.

(2) Let $f_1 = x^2+y+z+1, f_2 = x^2+y^2+3x, f_3 = y^2+xz+x+1, f_4 = y^2+y+x+5$
Show that when we regard f_1, f_2, f_3, f_4 as elements in $\mathbb{C}[x, y, z]$, then $\mathrm{V}(\langle f_1, f_2, f_3, f_4\rangle) = \{\}$, but if we instead regard them as elements in $\mathbb{Z}_2[x, y, z]$, then $\mathrm{V}(\langle f_1, f_2, f_3, f_4\rangle)$ is non-empty. Moreover, in the $\mathbb{Z}_2[x, y, z]$-case, you should determine $\mathrm{V}(\langle f_1, f_2, f_3, f_4\rangle)$ explicitly.

(3) In this exercise we will learn how to construct ideals with finite varieties. Consider the polynomial ring $k[x_1, \ldots, x_n]$, k a field. The equation $\mathrm{V}(I) = \{(a_1, \ldots, a_n)\}$ has the obvious solution $I = \langle x_1 - a_1, \ldots, x_n - a_n\rangle$, that is, $\mathrm{V}(\langle x_1 - a_1, \ldots, x_n - a_n\rangle) = \{(a_1, \ldots, a_n)\}$.

This gives a way to construct ideals with singleton varieties. What about a variety with two elements, say $\{(a_1, \ldots, a_n), (b_1, \ldots, b_n)\}$? Well, we have

$$\{(a_1, \ldots, a_n), (b_1, \ldots, b_n)\} = \{(a_1, \ldots, a_n)\} \cup \{(b_1, \ldots, b_n)\} =$$

$$\mathrm{V}(\langle x_1 - a_1, \ldots, x_n - a_n\rangle) \cup \mathrm{V}(\langle x_1 - b_1, \ldots, x_n - b_n\rangle)$$

and since $\mathrm{V}(I) \cup \mathrm{V}(J) = \mathrm{V}(I \cap J)$, this gives

$$\{(a_1, \ldots, a_n), (b_1, \ldots, b_n)\} =$$

$$\mathrm{V}(\langle x_1 - a_1, \ldots, x_n - a_n\rangle \cap \langle x_1 - b_1, \ldots, x_n - b_n\rangle).$$

This process may be continued, so to construct an ideal whose solution set is $\{p_1, \ldots, p_m\}$, each p_i being a point in k^n, we compute $I = I_1 \cap \cdots \cap I_m$, where $I_j = \langle x_1 - p_{j1}, \ldots, x_n - p_{jn}\rangle$.

The ideal I is called the *vanishing ideal* with respect to $p_1, \ldots, p_m$.

Let $R = ZZ/5[x, y, z]$.

(a) Let $p_1 = (1, 1, 1), p_2 = (0, 1, 1), p_3 = (0, 1, 2), p_4 = (0, 0, 3)$. Compute the vanishing ideal with respect to p_1, p_2, p_3. Call this ideal I. Then compute the vanishing ideal with respect to p_2, p_3, p_4. Call this ideal J.

(b) Compute a Gröbner basis for $I + J$. Check, by computing a GB and looking at the elements in the GB, that the corresponding variety equals $\{p_2, p_3\} (= \{p_1, p_2, p_3\} \cap \{p_2, p_3, p_4\})$.

(c) Compute a Gröbner basis for $I \cap J$. Check, by computing a GB and looking at the elements in the GB, that the corresponding variety equals $\{p_1, p_2, p_3, p_4\} (= \{p_1, p_2, p_3\} \cup \{p_2, p_3, p_4\})$.

(d) Compute a Gröbner basis for $I : J$. Check, by computing a GB and looking at the elements in the GB, that the corresponding variety equals $\{p_1\} (= \{p_1, p_2, p_3\} \setminus \{p_2, p_3, p_4\})$.
(e) Finally, compute a Gröbner basis for $I \cdot J$. Does it hold that $I \cdot J = I \cap J$? Now compute the radical ideal $\sqrt{I \cdot J}$ of $I \cdot J$. Check that $\sqrt{I \cdot J} = I \cap J$ holds.

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